# Applied Econometric Analysis using Logistic Regression (Stata)

# The Role of Digital Access, Economic & Demographic Indicators in Borrowing Decisions

Abhinayaa K Subramanian

## 1. Introduction:

Access to formal financial services remains uneven across nations, despite sustained efforts to expand financial inclusion. In particular, disparities in formal credit access limit people's ability to smooth consumption, invest lucratively, and manage financial risk.

This project investigates the relationship between digital access, education and formal borrowing behaviour using individual-level survey data. A logistic regression framework is employed to model the likelihood of formal borrowing as a function of internet access, digital payment usage, education, income, and other demographic controls.

Results indicate that digital access and higher education levels are positively associated with formal borrowing, even after controlling for income and other factors. Marginal effects suggest meaningful differences in borrowing probabilities across education, income and digital access groups, underscoring the significance of digital infrastructure and human capital in shaping financial inclusion outcomes.

## 2. Data & Variables

**2.1 Dataset:** World Bank's Global Findex Database 2021
The analysis uses data from the Global Findex database (World Bank Group 2021), a nationally representative, demand-side data source covering nearly 145,000 people in 139 economies, representing 97% of the world's population. It includes variables on individuals' borrowing behaviour, digital access, education level, income category and other demographic indicators, only from the year 2021 and restricting observations from all other years for consistency.

## 2.2 Dependent Variable

**Formal borrowing:** To study whether digital financial services like UPI payments, wireless transfers and financial literacy facilitate a shift from informal borrowing to regulated formal credit, the source of borrowing has been taken as a binary indicator **:**

- ***formal_borr***
  **1** - Borrowed from formal sources such as financial institutions
  **0** - Borrowed from Informal savings clubs, family or friends.

## 2.3 Key Independent Variables

**Digital Financial Access:** The advent of the internet allows individuals to access online banking, digital money and other credit tools. Greater engagement with digital finance is expected to lower access barriers, boost creditworthiness and exposure to regulated finance. This, in turn, raises the likelihood of formal borrowing and reduces the reliance on informal credit sources. To analyse this, the following 4 binary variables have been taken as measures for digital inclusion:

- ***internetaccess***
  **1** - Has access to internet connectivity
  **0** - Does not have access to internet connectivity
- ***has_credit***
  **1** - Owns a credit card
  **0** - Does not own a credit card
- ***mob_borrow***
  **1** - Uses mobile money account to borrow money
  **0** - Does not use mobile money account to borrow money
- ***anydigpayment***
  **1** – Made or received any digital payment
  **0** – Did not make or receive any digital payment

**Education:** Education level determines the credibility of individuals, their financial literacy and ability to repay loans through improved income prospects. Hence, we include education in our analysis in the following manner:

- ***educ***
  **0** - Respondent has completed Primary /Secondary Schooling
  **1** - Respondent has completed Tertiary education or more

**Income Quintile:** Income level determines financial stability, stronger credit profiles and repayment potential. While high-income individuals prefer formal sources, low-income individuals often turn to informal sources due to a lack of collateral, credit history and awareness. Hence, we include income quintiles to analyse how different economic strata make borrowing decisions:

- ***inc_q.*** - Categorical variable with values = 1 if Poorest 20% ; = 2 if Second 20% ; = 3 if Middle 20% ; = 4 if Fourth 20% ; = 5 if Richest 20%

**Control Variables:** The regression analysis controls for individual & household characteristics that may influence borrowing behaviour. The following variables help isolate the association between digital access, education and formal borrowing holding all other facts constant (ceterus paribus):

- ***age & age_sq*** - Continuous variable (taken in both linear and squared form to capture the non-linear effect of age on borrowing behaviour)
- ***female*** - Binary variable with value = 1 if female; = 0 if male
- ***rural*** - Binary variable with value = 1 if living in rural area; = 0 if living in urban area
- ***emp_in.*** - Binary variable with value = 1 if in workforce; = 0 if not in workforce

## 3. Research Framework

This paper employs a Logit regression model that approximates the probability of a person borrowing from official credit sources given a range of digital access, education and other demographic indicators. The logit model is appropriate here, as it allows for the estimation of the likelihood of formal borrowing while ensuring that predicted probabilities lie between zero and one. The model is estimated with robust standard errors to control for possible heteroskedasticity, and it is interpreted through marginal effects. The regression model is given as:

$$\mathbf{P\,(formal_borr_i = 1) = F\,\{\,\beta_0 + \beta_1\,internet_access_i + \beta_2\,mob_borrow_i + \beta_3\,anydigpayment_i + \beta_4\,has_credit_i + \beta_5\,educ_i + \beta_6\,inc_q_i + \beta_7\,age_i + \beta_8\,age_sq_i + \beta_9\,female_i + \beta_{10}\,rural_i + \beta_{11}emp_in_i\}}$$

## 3.1 Hypothesis Testing

Based on the logistic model above, the study tests the following hypotheses to analyse the determinants of formal borrowing behaviour:

### Hypothesis 1 : Digital Financial Access

**$H0_1$:** Digital access indicators (internet access, digital payment usage, mobile borrowing, and credit card ownership) have no significant effect on the likelihood of formal borrowing.

**$H1_1$:** Digital access indicators have a significant effect on the likelihood of formal borrowing.

### Hypothesis 2 : Economic Indicators

**$H0_2$:** Economic characteristics, including education level and income group, have no significant effect on the likelihood of formal borrowing.

**$H1_2$:** Economic characteristics have a significant effect on the likelihood of formal borrowing.

### Hypothesis 3 : Demographic Indicators

**$H0_3$:** Demographic Indicators (age, gender, rural residence and employment status) have no significant effect on the likelihood of formal borrowing.

**$H1_3$:** Demographic Indicators have a significant effect on the likelihood of formal borrowing.

## 4. Key Findings & Analysis

## 4.1 Model Significance and Fit (Using Table 1)

The log pseudolikelihood of –6863.67 indicates the model's overall fit to the data. The Wald chi-square statistic, $\chi^2 = 839.78$ ($p < 0.001$) confirms that the set of explanatory variables is jointly significant. The pseudo-$R^2$ of 0.0635 is consistent and suggests that while the model explains meaningful variation in formal borrowing behaviour, individual credit decisions are also influenced by unobserved factors.

**Table 1 : Logit Regression Output**

Logistic regression — Number of obs = 10,836; Wald chi2(14) = 839.78; Prob > chi2 = 0.0000; Log pseudolikelihood = -6863.6693; Pseudo R2 = 0.0635

| formal_borr | Coefficient | Robust std. err. | z | P>\|z\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| internetaccess | .3167775 | .0533938 | 5.93 | 0.000 | .2121276 | .4214275 |
| mob_borrow | .1765875 | .0865389 | 2.04 | 0.041 | .0069745 | .3462006 |
| anydigpayment | .3299709 | .0453926 | 7.27 | 0.000 | .241003 | .4189389 |
| has_credit | .9499163 | .0546128 | 17.39 | 0.000 | .8428771 | 1.056955 |
| 1.educ | .1404902 | .0546425 | 2.57 | 0.010 | .0333929 | .2475874 |
| inc_q | | | | | | |
| Second 20% | .289342 | .0797755 | 3.63 | 0.000 | .132985 | .445699 |
| Middle 20% | .2292377 | .0761422 | 3.01 | 0.003 | .0800016 | .3784738 |
| Fourth 20% | .2303654 | .074431 | 3.10 | 0.002 | .0844833 | .3762475 |
| Richest 20% | .2678074 | .0711053 | 3.77 | 0.000 | .1284437 | .4071712 |
| age | .0561316 | .0080136 | 7.00 | 0.000 | .0404251 | .071838 |
| c.age#c.age | -.0003978 | .0000923 | -4.31 | 0.000 | -.0005787 | -.0002168 |
| female | -.164528 | .0418192 | -3.93 | 0.000 | -.2464922 | -.0825638 |
| rural | .216166 | .0433265 | 4.99 | 0.000 | .1312477 | .3010843 |
| emp_in | -.1992693 | .0530189 | -3.76 | 0.000 | -.3031844 | -.0953542 |
| _cons | -2.416662 | .2200368 | -10.98 | 0.000 | -2.847926 | -1.985397 |

## 4.2 Margin Effects (Using Table 2)

### 4.2.1 Digital access strongly increases formal borrowing [ Hypothesis 1]

- Access to the internet increases the probability of formal borrowing by **7.0 percentage points** , holding other factors constant. *(p < 0.01)*
- Use of digital payment platforms is associated with a **7.3 percentage point** higher likelihood of formal borrowing. *(p < 0.01)*
- Mobile-based borrowing shows a smaller but statistically significant effect of **3.9 percentage point** increase. *(p < 0.05)*
- Credit card ownership shows the maximum increase in probability of formal credit of **21.02 percentage points**, among any other digital access indicators, highlighting the role of existing financial inclusion in furthering access to formal credit. *(p < 0.01)*

### 4.2.2 Economic indicators like education & income quintiles [ Hypothesis 2]

- Higher education is associated with a **3.14 percentage point** increase in likelihood of borrowing from formal credit sources, even after controlling for income & other factors.

- Individuals from higher income levels are **5 to 6 percentage points** more likely to use formal sources of borrowing when compared to the poorest 20% of the population, with the second 20% and the richest 20% being most likely. *(p < 0.01)*

### 4.2.3 Demographic indicators like age, gender, rural residence & work status [ Hypothesis 3]

- Age exhibits a non-linear relationship with formal borrowing. While the average marginal effect of age is positive, the negative coefficient on the age-squared term indicates a **diminishing marginal increase** in the likelihood of formal credit at higher ages.
- Women are **3.64 percentage points** less likely to borrow from formal sources than men.
- Conditional on income, education and digital access, rural residents are **4.78 percentage points** more likely to formally borrow than urban dwellers. This is probably due to targeted rural credit initiatives and their after-effects.
- Being employed in the workforce is associated with a **4.41 percentage points** decrease in the probability of formal borrowing, compared to unemployed individuals, which may be due to lesser credit needs owing to more stable income streams.
- The margin effects of all control variables in the model are **significant at the 1% level**.

**Table 2 : Margin Effects of Logit Regression Model**

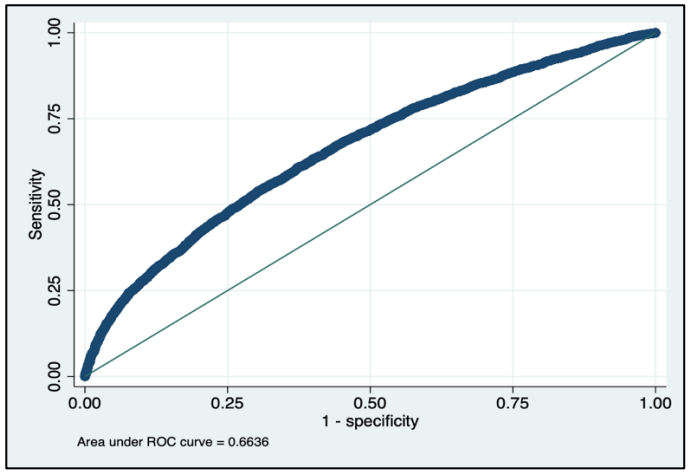
```
. margins, dydx(*) post

Average marginal effects                                Number of obs = 10,836
Model VCE: Robust

Expression: Pr(formal_borr), predict()
dy/dx wrt:  internetaccess mob_borrow anydigpayment has_credit 1.educ 2.inc_q
            3.inc_q 4.inc_q 5.inc_q age female rural emp_in
```

| | dy/dx | Delta-method std. err. | z | P>\|z\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| internetaccess | .0701025 | .0117393 | 5.97 | 0.000 | .0470938 | .0931112 |
| mob_borrow | .0390786 | .0191408 | 2.04 | 0.041 | .0015634 | .0765939 |
| anydigpayment | .0730222 | .0099555 | 7.33 | 0.000 | .0535099 | .0925346 |
| has_credit | .2102155 | .0114394 | 18.38 | 0.000 | .1877947 | .2326363 |
| 1.educ | .0314092 | .0123233 | 2.55 | 0.011 | .0072561 | .0555624 |
| inc_q | | | | | | |
| Second 20% | .0632223 | .0173532 | 3.64 | 0.000 | .0292106 | .0972339 |
| Middle 20% | .0498059 | .0164466 | 3.03 | 0.002 | .0175712 | .0820405 |
| Fourth 20% | .0500564 | .0160517 | 3.12 | 0.002 | .0185956 | .0815172 |
| Richest 20% | .0584011 | .0153222 | 3.81 | 0.000 | .02837 | .0884321 |
| age | .0057373 | .0003684 | 15.57 | 0.000 | .0050153 | .0064593 |
| female | -.0364099 | .0092303 | -3.94 | 0.000 | -.054501 | -.0183187 |
| rural | .0478373 | .0095505 | 5.01 | 0.000 | .0291186 | .066556 |
| emp_in | -.0440981 | .0117074 | -3.77 | 0.000 | -.0670442 | -.021152 |

Note: dy/dx for factor levels is the discrete change from the base level.

## 4.3 Measuring Goodness-of-Fit

### 4.3.1 ROC Curve Plot




The ROC curve yields an **AUC of 0.664**, indicating moderate discriminatory power. This suggests that the model can correctly distinguish a formal borrower from an informal borrower around 66% of the time, substantially better than random classification.

### 4.3.2 Hosmer - Lemeshow Test

```
Goodness-of-fit test after logistic model
Variable: formal_borr

 Number of observations = 10,836
       Number of groups =     10
Hosmer-Lemeshow chi2(8) =  10.73
            Prob > chi2 = 0.2174
```

The Hosmer – Lemeshow goodness-of- fit test has a **p-value = 0.2174** (> 0.05), indicating that the logit model fits the data well and does not significantly deviate from observed outcomes.

## 5. Policy Implications

- **Improving access to digital infrastructure**
  Expanding digital access can directly propel formal borrowing behaviour, even after controlling for education and income levels.

- **Strengthening financial touchpoints**
  The strong effect of credit card ownership indicates that once the access barrier is crossed, this early engagement acts as a gateway to broader formal credit sources.

- **Expanding digital financial literacy**
  Lack of awareness and trust hinders the use of formal credit tools, which can be avoided by incorporating financial literacy modules through school curricula, campaigns and mobile channels.

- **Designing gender-inclusive financial products**
  Since women are less likely to avail formal credit than men, initiatives like collateral-free microloans to women's self-help groups could meaningfully boost financial inclusion.

- **Targeting interventions by demographics**
  Heterogeneity across income and demographic groups indicate that targeted interventions work better than uniform credit expansion policies.

## 6. Limitations

- The analysis is based on cross-sectional survey data and identifies associations rather than causal effects between digital access and formal borrowing.
- Formal borrowing behaviour is self-reported & may be subject to reporting or recall bias.
- Although the model includes key factors affecting formal credit, other factors like psychological biases and cultural preferences have not been added, raising the possibility of omitted variable bias.
- Informal and cash-dependent borrowers may be underrepresented in the survey data, potentially underestimating the issues faced by financially underserved populations.

## 7. Conclusion

Using household-level survey data, this project applies a logit model with robust measures to analyse the factors involved in formal borrowing behaviour. Results indicate that digital access, especially internet access and digital payment methods, significantly increases the probability of borrowing from formal credit sources, even after controlling for income and education. Credit card ownership is the strongest predictor of formal borrowing, while gaps are prevalent among different demographic groups. Education and income levels are positively associated with the likelihood of formal credit usage.

Model diagnostics of ROC with AUC = 0.664 and the Hosmer-Lemeshow test indicate adequate predictive power. Overall, the findings indicate that digital access has a significant, positive impact on formal financial inclusion. However, targeted policy and credit mechanisms are still crucial to address structural barriers in achieving this.

[ Detailed Stata codebook available upon request.]